Indefinite integrals

1 Consider the integral $\int 5\left(5x+4\right)^3 dx$ and the substitution $u = 5x+4$.

a Find $\dfrac{du}{dx}$.

b Hence, find $\int 5\left(5x+4\right)^3 dx$.

2 Find $\int 2x\left(x^2-5\right)^7 dx$ using the substitution $u = x^2-5$.

3 Find $\int 3x^2\left(x^3+7\right)^4 dx$ using the substitution $u = x^3+7$.

4 Find $\int \dfrac{6x}{\left(3x^2+2\right)^2} dx$ using substitution.

Definite integrals

5 Consider the definite integral $\int_0^2 3\left(3x-6\right)^2 dx$ and the substitution $u = 3x-6$.

a Find the value of u when the value of x is:

i $x = 0$

ii $x = 2$

b Find $\dfrac{du}{dx}$.

c What does the integral, $\int_0^2 3\left(3x-6\right)^2 dx$, become after making the substitution $u = 3x-6$?

d Evaluate this integral.

6 Find $\int_0^3 9\left(x+3\right)^2 dx$ using the substitution $u = x+3$.

7 Find $\int_6^8 \left(3x-15\right)^2 dx$ using the substitution $u = 3x-15$.

8 Find $\int_1^2 4x\left(2x^2-2\right)^2 dx$ using the substitution $u = 2x^2-2$.

9 Find $\int_{-1}^2 64x\left(4x^2+1\right)^3 dx$ using the substitution $u = 4x^2+1$.

10 Find $\int_0^1 \dfrac{-144x^3}{\left(2x^4-3\right)^3} dx$ using the substitution $u = 2x^4-3$.

11 Find $\int_{-2}^1 \dfrac{24}{\left(\frac{x+5}{3}\right)^4} dx$ using the substitution $u = \dfrac{x+5}{3}$.

12 Find the following definite integrals using substitution:

a $\int_{-1}^{2} 3x^2\left(x^3-2\right)^2 dx$

b $\int_{-1}^{2} -81x^2\left(3x^3-8\right)^2 dx$

c $\int_{0}^{2} 24x\left(4x^2+9\right)^{0.5} dx$

d $\int_{0}^{3} 16x\sqrt{8x^2+9}dx$

e $\int_{0}^{1} 24x^3\left(3x^4+3\right)^2 dx$

f $\int_{17}^{53} 9\sqrt{3x-15}dx$

g $\int_{-1}^{3} 8\sqrt[3]{2x+2}dx$

h $\int_{0}^{1} \frac{48x}{\left(3x^2-1\right)^3}dx$

i $\int_{-2}^{1} \frac{-216x^2}{\left(x^3+2\right)^3}dx$

j $\int_{1}^{2} \frac{50x}{\sqrt{5x^2-4}}dx$

k $\int_{-1}^{2} (2x+5)\left(x^2+5x-8\right)^3 dx$

l $\int_{0}^{1} \left(4x^3+5\right)\left(x^4+5x+2\right)^3 dx$

m $\int_{13}^{18} x\sqrt{x-9}dx$